Annex II. Debt Sustainability Analysis Update

Rwanda: Risk Rating Summary	
Joint Bank-Fund Debt Sustainability Analysis [1]	
Risk of external debt distress	Moderate [2]
Overall risk of debt distress	Moderate
Granularity in the risk rating	Limited space to absorb shocks
Application of judgment	No
Macroeconomic projections	The DSA incorporates the macroeconomic projections in the 4th PCI review staff report. Due to the prolonged pandemic, the output recovery is expected to be slower than in the December 2020 DSA, despite sizable fiscal support. The medium-term fiscal consolidation path is envisaged after the pandemic abates.
Financing strategy	As in the December 2020 DSA, the current DSA assumes continued support from bilateral and multilateral development partners over the medium term. From 2028 onwards, the financing mix is assumed to: (i) shift gradually away from concessional financing to market-based financing as Rwanda's income level rises, and (ii) shift from external to domestic financing as the local bond markets develop. As of May 25, 2021, the World Bank disbursed US$513.5 million in FY21 in the form of concessional financing, including budget support (loans and grants) and project financing (loans and grants).
Realism tools flagged	The envisaged post-pandemic fiscal adjustment is assessed as ambitious based on the experience in LICs.
Mechanical risk rating under the external DSA	Moderate

[1] This debt sustainability analysis was conducted using the Joint Bank-Fund Debt Sustainability Framework for Low-Income Countries (LIC-DSF) that was approved in 2017. The fiscal year for Rwanda is from July–June; however, this DSA is prepared on a calendar year basis.

[2] Rwanda's Composite Indicator score (3.16) has not been updated since the December 2020 DSA. Therefore, Rwanda's debt-carrying capacity remains unchanged, and assessed as strong. Debt coverage includes the central government and state-owned enterprises (SOEs). There has been no change in debt coverage since the December 2020 DSA.

The updated Bank-Fund assessment of Rwanda's debt sustainability indicates continued moderate risk of external and overall debt distress, although it also suggests elevated solvency risks and limited buffers to cushion the impact of external and growth shocks.

Total nominal public debt-to-GDP stood at 71.3 percent in 2020, which is higher than expected compared to the December 2020 DSA against the background of lower-than-expected GDP and a higher fiscal deficit due to the larger pandemic-related spending.

The baseline scenario in the updated DSA is consistent with the macroeconomic projections and the post-pandemic medium-term fiscal consolidation envisaged under the 4th PCI review. As the pandemic continues to impact Rwanda's economy, the output recovery is expected to be somewhat slower than in the December 2020 DSA, despite sizable fiscal support (Text Table below). Output is projected to grow by 5.1 percent in 2021, reflecting strong base effects and a pickup in the agriculture and manufacturing sectors, before faster broad-based recovery starts in 2022. Compared to the December 2020 DSA, the updated DSA assumes larger fiscal spending to continue responding to the health, social, and economic fallout of the pandemic, which is expected to contribute to a faster pace of accumulation of public and publicly guaranteed (PPG) debt over time. It also incorporates a debt management strategy which helps to mitigate the refinancing risk associated with the $400 million Eurobond bullet payment in 2023.

Key Macro and Debt Indicators—Comparison with the Previous DSA[1]

Calendar year	2020	2021	2026	2031	2036	2041	2026-41
			Projections				
Selected indicators from the macro-framework and debt data (Percent, unless otherwise indicated)							
			PV of PPG External Debt to GDP Ratio				
2020 DSA	34.1	35.2	38.9	36.6	30.9	...	33.7
2021 DSA	33.9	38.4	43.0	36.7	31.9	25.4	34.5
			PV of Public Debt to GDP Ratio				
2020 DSA	45.5	48.8	47.7	44.8	47.1	...	46.4
2021 DSA	51.0	55.7	51.9	44.8	44.7	45.9	45.8
			Grant Element of New External Borrowing				
2020 DSA	48.3	47.5	44.3	40.3	31.4	...	35.5
2021 DSA	-	30.3	38.0	31.6	30.1	18.7	32.6
			Stock of New Commercial Loan (billions of U.S. dollars)				
2020 DSA	0.0	0.0	0.9	1.0	1.1	...	1.1
2021 DSA	0.0	0.5	0.9	0.6	1.4	2.6	1.3
			Real GDP Growth (annual percent change)				
2020 DSA	-0.2	5.7	6.1	7.2	6.8	...	6.8
2021 DSA	-3.4	5.1	6.1	7.2	6.8	6.5	6.8
			Current Account Balance (percent of GDP)				
2020 DSA	-12.2	-12.5	-7.6	-8.0	-7.5	...	-7.5
2021 DSA	-12.2	-13.4	-6.8	-8.0	-7.5	-6.5	-7.4
			Exports of goods and services (percent of GDP)				
2020 DSA	18.2	22.7	28.5	30.3	32.2	...	31.2
2021 DSA	18.7	22.3	30.8	29.6	28.3	27.5	29.0
			Fiscal balance[2] (percent of GDP)				
2020 DSA	-9.7	-7.9	-3.7	-5.2	-4.8	...	-4.8
2021 DSA	-9.4	-8.9	-3.4	-4.1	-3.4	-3.3	-3.6

1/ Previous DSA completed in December 2020 (published in January 2021).
2/ Fiscal balance excludes debt assumption for Marriott loan.
Sources: Rwandan authorities; IMF and World Bank staff estimates and projections.

In the baseline scenario, the present value (PV) of PPG debt is projected to reach 56.7 percent of GDP in 2022 before achieving the East African Community debt convergence criterion of 50 percent in 2027. The nominal PPG debt-to-GDP ratio is projected to peak at 81.3 percent in 2022 and reach the nominal debt anchor of 65 percent under the PCI program in 2030, two years later than expected in the December 2020 DSA.[1] This would require a cumulative reduction in the overall headline deficit of 6 percentage points of GDP between 2021 and 2029, of which about half is due to the gradual phasing-out of COVID-19-related spending (with no additional fiscal effort or drag on growth), and the other half through the implementation of additional domestic revenue mobilization and spending rationalization measures with the support of ongoing IMF TA, the bulk of which will be implemented after growth returns to pre-pandemic levels in 2023.

The stress tests indicate that Rwanda's solvency risks are elevated overall, which makes the country more susceptible to both external and growth shocks compared to the December 2020 DSA (Figure 1 and 2). Solvency indicators for the PV of external debt-to-GDP ratio and PV of total public debt-to-GDP ratio are below the indicative threshold/benchmark under the baseline scenario, but there are multiple breaches under the most extreme risk shocks. On the other hand, liquidity indicators (debt service-to-exports ratio and debt service-to-revenue ratio) show that the liquidity risks associated with the repayment of the existing Eurobond in 2023 are somewhat mitigated by the debt management strategy to smooth out the debt servicing profile. A temporary deterioration of liquidity indicators in 2021 mainly reflects the recording of the debt service smoothing operations. Rwanda may still face liquidity pressures due to adverse market conditions, as pointed to by a breach of the market-financing risk indicator for gross financing needs (Figure 6) but this is mitigated by adequate reserves and available external official financing.

Overall, while liquidity indicators display only one short-lived breach in 2021 under the baseline scenario, multiple breaches in both solvency and liquidity risk indicators under the most extreme shocks indicate a moderate risk of debt distress. Furthermore, a granular assessment of the moderate risk rating shows that Rwanda has limited space to absorb shocks (Figure 5). To mitigate these risks, the government needs to adopt a credible fiscal consolidation path as soon as the COVID-19 crisis abates and accelerate efforts to strengthen fiscal risk management.

The evolution of public debt going forward is dominated by the path of the primary fiscal deficit and real GDP growth, as in the past (Figure 3). Compared to the past five years, a higher primary deficit due to the COVID-19 shock is expected to raise public debt regardless of growth recovery. Realism tools illustrate that the projected 3-year fiscal adjustment peaking at 3.8 percentage points of GDP lies in the top quartile of the distribution of past adjustment for a sample of LICs, signaling that the envisaged post-pandemic fiscal adjustment in the baseline scenario is ambitious based on the experience in LICs (Figure 4).

The current macroeconomic framework, which underpins this DSA, reflects currently available information. However, updates with respect to the economic impact and policy response to the COVID-19 crisis are evolving and risks to the debt outlook and sustainability could elevate should the pandemic shock prolong further.

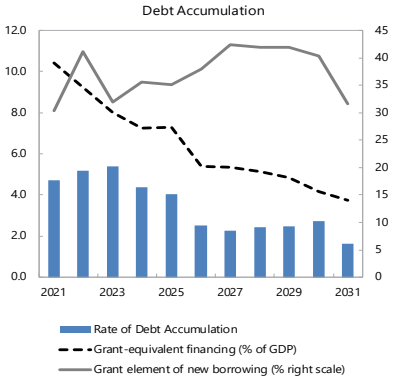
[1] See IMF Country Report No. 2021/001

Table 1. Rwanda: External Debt Sustainability Framework, Baseline Scenario, 2020–41

(In percent of GDP, unless otherwise indicated)

	Actual	Projections								Average 8/	
	2020	2021	2022	2023	2024	2025	2026	2031	2041	Historical	Projections
External debt (nominal) 1/	**63.8**	**71.9**	**76.2**	**78.6**	**79.1**	**78.9**	**77.5**	**70.3**	**58.2**	**39.8**	**75.4**
of which: public and publicly guaranteed (PPG)	**55.3**	**63.0**	**66.8**	**68.7**	**68.7**	**67.9**	**65.9**	**55.8**	**35.1**	**33.8**	**63.8**
Change in external debt	5.3	8.1	4.3	2.4	0.5	-0.2	-1.4	-2.0	-1.2		
Identified net debt-creating flows	**11.4**	**6.3**	**3.7**	**2.0**	**0.9**	**-1.1**	**-0.1**	**0.0**	**-2.1**	**6.6**	**1.0**
Non-interest current account deficit	**10.9**	**12.0**	**10.7**	**9.7**	**7.9**	**5.7**	**5.0**	**6.2**	**4.2**	**9.7**	**7.3**
Deficit in balance of goods and services	16.4	18.9	17.6	16.2	14.7	12.7	10.9	10.0	5.0	**15.7**	**13.0**
Exports	18.7	22.3	25.7	27.7	29.5	31.0	30.8	29.6	27.5		
Imports	35.1	41.2	43.4	43.9	44.3	43.7	41.7	39.6	32.4		
Net current transfers (negative = inflow)	-6.0	-7.9	-8.0	-7.7	-8.0	-7.9	-6.6	-4.8	-2.0	**-7.4**	**-6.7**
of which: official	-5.8	-7.8	-6.9	-6.3	-6.3	-6.4	-4.8	-2.9	-1.2		
Other current account flows (negative = net inflow)	0.5	1.0	1.1	1.2	1.1	0.9	0.7	0.9	1.2	**1.3**	**0.9**
Net FDI (negative = inflow)	**-1.0**	**-3.9**	**-3.8**	**-3.5**	**-3.1**	**-3.1**	**-2.6**	**-3.3**	**-5.0**	**-2.6**	**-3.2**
Endogenous debt dynamics 2/	**1.5**	**-1.8**	**-3.3**	**-4.2**	**-3.8**	**-3.7**	**-2.6**	**-2.9**	**-1.2**		
Contribution from nominal interest rate	1.3	1.4	1.5	1.5	1.7	1.8	1.8	1.8	2.3		
Contribution from real GDP growth	2.0	-3.2	-4.7	-5.7	-5.5	-5.4	-4.4	-4.7	-3.6		
Contribution from price and exchange rate changes	-1.8	...	...	...	...	...	...	...	...		
Residual 3/	**-6.1**	**1.8**	**0.6**	**0.4**	**-0.4**	**0.9**	**-1.3**	**-2.0**	**0.8**	**-1.9**	**-0.4**
of which: exceptional financing	-0.2	-0.4	-0.1	0.0	0.0	0.0	0.0	0.0	0.0		
Sustainability indicators											
PV of PPG external debt-to-GDP ratio	**33.9**	**38.4**	**41.0**	**42.9**	**43.8**	**44.0**	**43.0**	**36.7**	**25.4**		
PV of PPG external debt-to-exports ratio	**181.4**	**171.9**	**159.4**	**154.7**	**148.5**	**141.9**	**139.8**	**124.2**	**92.3**		
PPG debt service-to-exports ratio	**10.8**	**27.1**	**8.0**	**12.2**	**7.7**	**8.2**	**9.2**	**14.1**	**12.1**		
PPG debt service-to-revenue ratio	**10.8**	**31.9**	**10.9**	**17.7**	**11.6**	**12.5**	**13.5**	**19.0**	**13.0**		
Gross external financing need (Billion of U.S. dollars)	1.4	1.7	1.2	1.4	1.1	1.0	1.1	2.3	3.6		
Key macroeconomic assumptions											
Real GDP growth (in percent)	-3.4	5.1	7.0	8.1	7.5	7.5	6.1	7.2	6.5	**6.1**	**6.9**
GDP deflator in US dollar terms (change in percent)	3.2	-4.3	-0.8	0.1	0.3	1.2	2.0	2.0	2.0	**-0.6**	**0.8**
Effective interest rate (percent) 4/	2.3	2.3	2.2	2.1	2.3	2.4	2.5	2.7	4.3	**3.2**	**2.5**
Growth of exports of G&S (US dollar terms, in percent)	-14.4	20.3	22.4	16.5	14.8	14.3	7.3	7.8	8.1	**11.4**	**12.4**
Growth of imports of G&S (US dollar terms, in percent)	-3.0	18.2	11.8	9.4	8.8	7.5	3.1	6.7	7.1	**8.6**	**9.0**
Grant element of new public sector borrowing (in percent)	...	30.3	41.1	31.9	35.6	35.1	38.0	31.6	18.7	**...**	**37.3**
Government revenues (excluding grants, in percent of GDP)	18.8	19.0	18.9	19.2	19.5	20.3	21.0	21.9	25.6	**14.9**	**20.5**
Aid flows (in Billion of US dollars) 5/	0.5	1.2	1.1	1.0	1.0	1.1	0.9	1.0	0.8		
Grant-equivalent financing (in percent of GDP) 6/	...	10.4	9.2	8.0	7.3	7.3	5.4	3.7	1.3	**...**	**6.4**
Grant-equivalent financing (in percent of external financing) 6/	...	53.4	64.4	55.4	63.0	63.7	64.0	49.1	34.1	**...**	**60.3**
Nominal GDP (Billion of US dollars)	10	10	11	12	13	14	15	23	55		
Nominal dollar GDP growth	-0.2	0.6	6.2	8.2	7.9	8.8	8.2	9.3	8.6	**5.4**	**7.7**
Memorandum items:											
PV of external debt 7/	42.4	47.3	50.4	52.8	54.3	55.0	54.6	51.2	48.4		
In percent of exports	226.8	212.0	195.8	190.5	184.0	177.3	177.5	173.2	176.4		
Total external debt service-to-exports ratio	19.7	34.9	14.7	18.7	14.1	14.5	16.0	22.9	27.2		
PV of PPG external debt (in Billion of US dollars)	3.5	4.0	4.5	5.1	5.6	6.2	6.5	8.6	13.9		
(PVt-PVt-1)/GDPt-1 (in percent)		4.7	5.2	5.4	4.4	4.1	2.5	1.7	0.7		
Non-interest current account deficit that stabilizes debt ratio	5.6	3.8	6.5	7.3	7.3	5.9	6.4	8.2	5.4		

Definition of external/domestic debt	Currency-based
Is there a material difference between the two criteria?	No



Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Current-year interest payments divided by previous period debt stock.

5/ Defined as grants, concessional loans, and debt relief.

6/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

7/ Assumes that PV of private sector debt is equivalent to its face value.

8/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

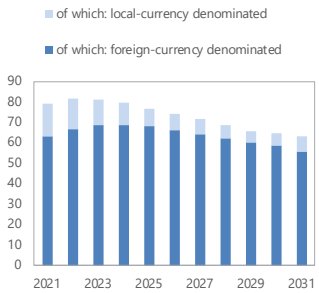
Table 2. Rwanda: Public Sector Debt Sustainability Framework, Baseline Scenario, 2020–41

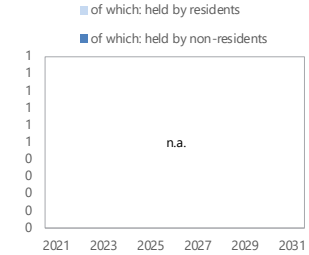
(In percent of GDP, unless otherwise indicated)

	Actual	Projections								Average 6/	
	2020	2021	2022	2023	2024	2025	2026	2031	2041	Historical	Projections
Public sector debt 1/	**71.3**	**79.1**	**81.3**	**81.1**	**79.5**	**76.5**	**74.0**	**63.2**	**55.2**	**43.1**	**73.2**
of which: external debt	**55.3**	**63.0**	**66.8**	**68.7**	**68.7**	**67.9**	**65.9**	**55.8**	**35.1**	**33.8**	**63.8**
Change in public sector debt	8.5	7.8	2.2	-0.2	-1.7	-3.0	-2.5	-1.2	-0.1		
Identified debt-creating flows	**10.1**	**7.9**	**2.4**	**0.1**	**-1.5**	**-2.7**	**-2.3**	**-1.2**	**-0.1**	**1.9**	**-0.6**
Primary deficit	**8.7**	**6.9**	**5.6**	**4.9**	**3.4**	**2.2**	**1.9**	**3.1**	**2.5**	**3.1**	**3.3**
Revenue and grants	23.6	25.5	24.6	24.2	24.5	25.4	24.5	23.9	26.3	**21.3**	**24.5**
of which: grants	4.8	6.5	5.7	5.0	4.9	5.0	3.5	2.0	0.8		
Primary (noninterest) expenditure	32.2	32.4	30.2	29.1	27.8	27.6	26.4	27.0	28.8	**24.4**	**27.8**
Automatic debt dynamics	**1.4**	**1.0**	**-3.2**	**-4.8**	**-4.7**	**-4.8**	**-4.0**	**-4.2**	**-2.6**		
Contribution from interest rate/growth differential	1.9	-2.7	-4.6	-5.5	-5.4	-5.0	-3.9	-4.3	-2.6		
of which: contribution from average real interest rate	-0.3	0.8	0.6	0.6	0.3	0.5	0.4	0.0	0.8		
of which: contribution from real GDP growth	2.2	-3.5	-5.2	-6.1	-5.7	-5.5	-4.4	-4.3	-3.4		
Contribution from real exchange rate depreciation	-0.4	...	...	...	...	...	...	...	...		
Other identified debt-creating flows	**0.0**	**0.0**	**0.0**	**0.0**	**-0.2**	**-0.1**	**-0.1**	**0.0**	**0.0**	**-0.1**	**-0.1**
Privatization receipts (negative)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Recognition of contingent liabilities (e.g., bank recapitalization)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Debt relief (HIPC and other)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Other debt creating or reducing flow (use of earmarked fund)	0.0	0.0	0.0	0.0	-0.2	-0.1	-0.1	0.0	0.0		
Residual	**-1.6**	**3.5**	**1.2**	**0.4**	**0.6**	**0.0**	**-0.4**	**0.0**	**0.0**	**3.3**	**0.4**
Sustainability indicators											
PV of public debt-to-GDP ratio 2/	**51.0**	**55.7**	**56.7**	**56.5**	**55.5**	**53.4**	**51.9**	**44.8**	**45.9**		
PV of public debt-to-revenue and grants ratio	**216.4**	**218.6**	**230.6**	**233.5**	**227.1**	**210.7**	**211.7**	**187.4**	**174.5**		
Debt service-to-revenue and grants ratio 3/	**36.5**	**55.9**	**36.8**	**41.6**	**33.0**	**30.2**	**29.8**	**32.2**	**39.2**		
Gross financing need 4/	17.3	21.1	14.7	14.9	11.3	9.7	9.1	10.7	12.8		
Key macroeconomic and fiscal assumptions											
Real GDP growth (in percent)	-3.4	5.1	7.0	8.1	7.5	7.5	6.1	7.2	6.5	**6.1**	**6.9**
Average nominal interest rate on external debt (in percent)	1.4	1.4	1.4	1.4	1.6	1.6	1.7	1.5	2.1	**1.8**	**1.5**
Average real interest rate on domestic debt (in percent)	-0.7	8.7	5.4	4.6	5.8	6.2	6.6	5.4	4.3	**1.3**	**6.3**
Real exchange rate depreciation (in percent, + indicates depreciation)	-0.8	...	...	...	...	...	...	...	...	**2.4**	**...**
Inflation rate (GDP deflator, in percent)	8.3	1.9	5.4	5.8	5.0	5.0	5.0	5.0	5.0	**4.3**	**4.8**
Growth of real primary spending (deflated by GDP deflator, in percent)	28.2	5.7	-0.1	3.9	2.9	6.4	1.5	7.0	8.9	**11.1**	**5.2**
Primary deficit that stabilizes the debt-to-GDP ratio 5/	0.2	-0.9	3.4	5.0	5.0	5.2	4.4	4.2	2.6	**-1.5**	**4.1**
PV of contingent liabilities (not included in public sector debt)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		

Definition of external/domestic debt	**Currency-based**
Is there a material difference between the two criteria?	**No**

Public sector debt 1/





Sources: Country authorities; and staff estimates and projections.

1/ Coverage of debt: The central government plus social security and extra budgetary funds, central bank, government-guaranteed debt, non-guaranteed SOE debt. Definition of external debt is Currency-based.

2/ The underlying PV of external debt-to-GDP ratio under the public DSA differs from the external DSA with the size of differences depending on exchange rates projections.

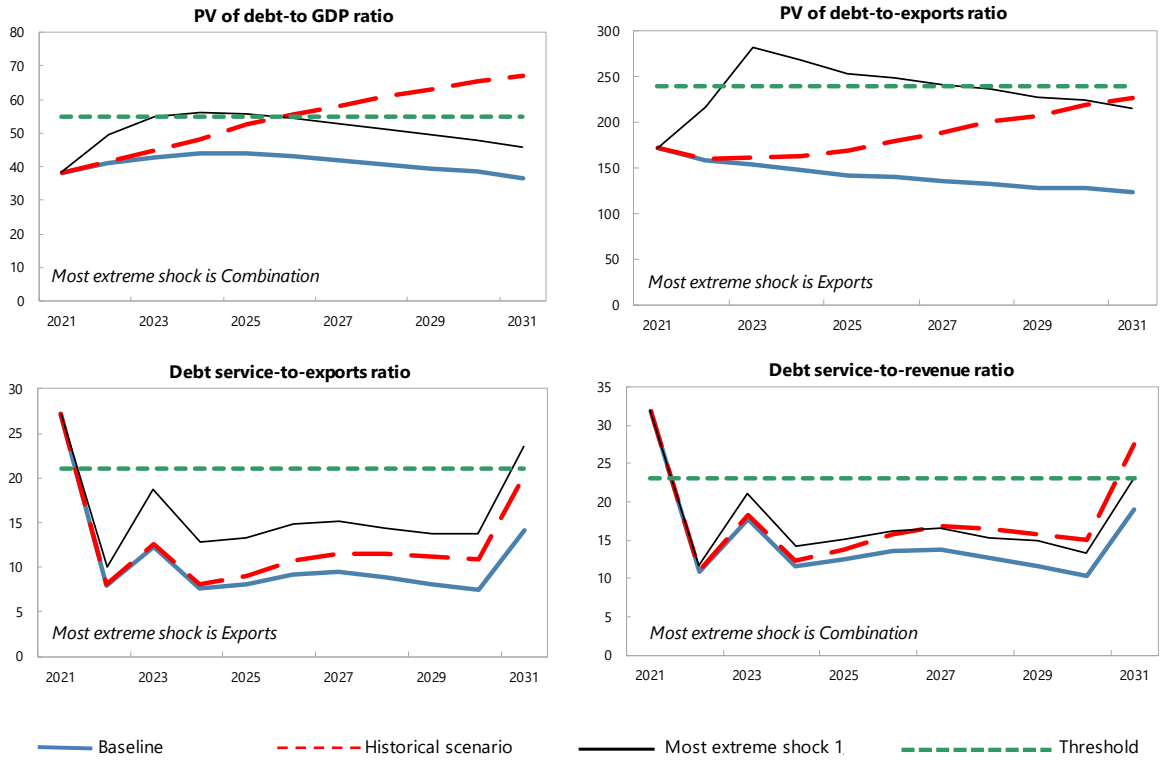
3/ Debt service is defined as the sum of interest and amortization of medium and long-term, and short-term debt.

4/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period and other debt creating/reducing flows.

5/ Defined as a primary deficit minus a change in the public debt-to-GDP ratio ((-): a primary surplus), which would stabilizes the debt ratio only in the year in question.

6/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

Figure 1. Rwanda: Indicators of Public and Publicly Guaranteed External Debt Under Alternative Scenarios, 2021–31 [1/ 2/]



Customization of Default Settings	Size	Interactions
Tailored Tests		
Combined CLs	No	
Natural Disasters	n.a.	n.a.
Commodity Prices [2/]	n.a.	n.a.
Market Financing	No	No

Note: "Yes" indicates any change to the size or interactions of the default settings for the stress tests. "n.a." indicates that the stress test does not apply.

Borrowing Assumptions for Stress Tests*	Default	User defined
Shares of marginal debt		
External PPG MLT debt	100%	
Terms of marginal debt		
Avg. nominal interest rate on new borrowing in USD	2.0%	2.0%
USD Discount rate	5.0%	5.0%
Avg. maturity (incl. grace period)	27	27
Avg. grace period	6	6

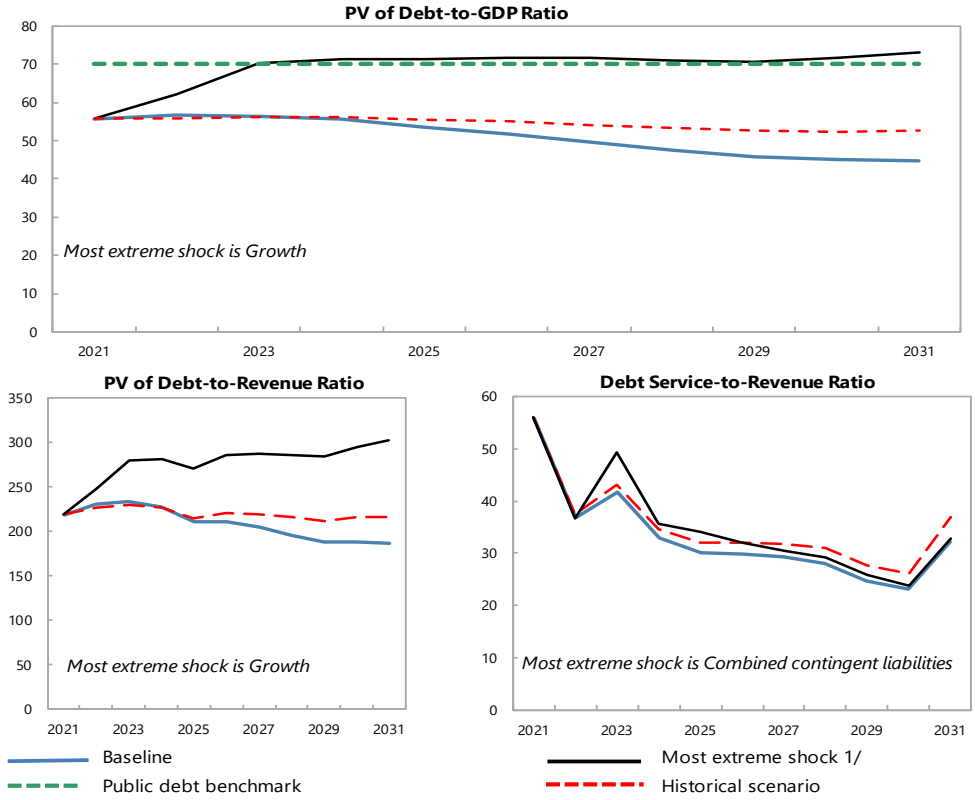
* Note: All the additional financing needs generated by the shocks under the stress tests are assumed to be covered by PPG external MLT debt in the external DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2031. Stress tests with one-off breaches are also presented (if any), while these one-off breaches are deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

2/ The magnitude of shocks used for the commodity price shock stress test are based on the commodity prices outlook prepared by the IMF research department.

Figure 2. Rwanda: Indicators of Public Debt Under Alternative Scenarios, 2021–31 1/



Borrowing Assumptions for Stress Tests*	Default	User defined
Shares of marginal debt		
External PPG medium and long-term	61%	61%
Domestic medium and long-term	14%	14%
Domestic short-term	26%	26%
Terms of marginal debt		
External MLT debt		
Avg. nominal interest rate on new borrowing in USD	2.0%	2.0%
Avg. maturity (incl. grace period)	27	27
Avg. grace period	6	6
Domestic MLT debt		
Avg. real interest rate on new borrowing	5.6%	5.6%
Avg. maturity (incl. grace period)	3	3
Avg. grace period	2	2
Domestic short-term debt		
Avg. real interest rate	1%	1.0%

* Note: The public DSA allows for domestic financing to cover the additional financing needs generated by the shocks under the stress tests in the public DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2031. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

Table 3. Rwanda: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2021–31

(In percent)

	Projections										
	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031
PV of debt-to GDP ratio											
Baseline	**38.4**	**41.0**	**42.9**	**43.8**	**44.0**	**43.0**	**41.8**	**40.6**	**39.3**	**38.5**	**36.7**
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2021-2041 1/	38.4	41.3	44.8	48.2	52.6	55.5	58.2	61.0	63.1	65.5	67.0
B. Bound Tests											
B1. Real GDP growth	38.4	44.2	50.6	51.7	51.9	50.7	49.3	47.8	46.4	45.4	43.3
B2. Primary balance	38.4	42.0	46.4	47.5	47.6	46.6	45.3	44.0	42.6	41.5	39.6
B3. Exports	38.4	44.7	52.8	53.4	53.2	51.9	50.3	48.7	46.9	45.3	42.9
B4. Other flows 2/	38.4	44.2	48.7	49.4	49.3	48.1	46.7	45.2	43.7	42.4	40.2
B6. One-time 30 percent nominal depreciation	38.4	52.0	49.9	51.2	51.7	50.6	49.2	47.8	46.4	45.6	43.7
B6. Combination of B1-B5	38.4	49.6	55.1	56.0	55.9	54.6	53.0	51.3	49.5	48.1	45.7
C. Tailored Tests											
C1. Combined contingent liabilities	38.4	43.9	46.4	47.4	47.8	46.8	45.5	44.1	42.8	41.8	39.9
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	38.4	45.9	48.0	49.2	49.3	48.1	46.6	45.1	43.7	42.8	40.8
Threshold	**55**	**55**	**55**	**55**	**55**	**55**	**55**	**55**	**55**	**55**	**55**
PV of debt-to-exports ratio											
Baseline	**171.9**	**159.4**	**154.7**	**148.5**	**141.9**	**139.8**	**135.6**	**133.3**	**128.8**	**128.4**	**124.2**
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2021-2041 1/	171.9	160.5	161.5	163.2	169.5	180.5	188.7	200.4	206.5	218.6	226.4
B. Bound Tests											
B1. Real GDP growth	171.9	159.4	154.7	148.5	141.9	139.8	135.6	133.3	128.8	128.4	124.2
B2. Primary balance	171.9	163.1	167.3	160.8	153.5	151.6	147.0	144.4	139.4	138.5	133.8
B3. Exports	171.9	216.5	282.0	267.9	253.9	249.5	241.5	236.8	227.5	224.0	214.8
B4. Other flows 2/	171.9	171.7	175.5	167.3	159.0	156.4	151.5	148.7	143.0	141.4	136.1
B6. One-time 30 percent nominal depreciation	171.9	159.4	141.8	136.8	131.3	129.5	125.7	123.7	119.8	120.1	116.6
B6. Combination of B1-B5	171.9	209.9	174.0	216.5	205.9	202.6	196.3	192.6	185.1	183.1	176.3
C. Tailored Tests											
C1. Combined contingent liabilities	171.9	170.5	167.2	160.5	154.0	152.1	147.6	145.1	140.2	139.4	134.9
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	171.9	159.4	154.8	148.9	142.2	139.9	135.2	132.4	128.1	127.6	123.3
Threshold	**240**	**240**	**240**	**240**	**240**	**240**	**240**	**240**	**240**	**240**	**240**
Debt service-to-exports ratio											
Baseline	**27.1**	**8.0**	**12.2**	**7.7**	**8.2**	**9.2**	**9.5**	**8.9**	**8.1**	**7.4**	**14.1**
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2021-2041 1/	27.1	8.1	12.6	8.2	9.0	10.7	11.6	11.5	11.2	10.9	20.4
B. Bound Tests											
B1. Real GDP growth	27.1	8.0	12.2	7.7	8.2	9.2	9.5	8.9	8.1	7.4	14.1
B2. Primary balance	27.1	8.0	12.5	8.3	8.7	9.8	10.0	9.4	8.7	8.1	14.8
B3. Exports	27.1	10.0	18.8	12.8	13.3	14.9	15.2	14.3	13.8	13.8	23.5
B4. Other flows 2/	27.1	8.0	12.6	8.2	8.6	9.7	9.9	9.3	8.9	8.5	15.1
B6. One-time 30 percent nominal depreciation	27.1	8.0	12.2	7.3	7.9	8.9	9.2	8.7	7.9	6.7	13.4
B6. Combination of B1-B5	27.1	9.4	16.6	10.8	11.3	12.7	13.0	12.2	12.0	11.0	19.6
C. Tailored Tests											
C1. Combined contingent liabilities	27.1	8.0	12.5	8.0	8.5	9.5	9.8	9.2	8.4	7.7	14.3
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	27.1	8.0	12.3	8.4	9.6	10.5	9.9	10.8	7.5	7.3	13.9
Threshold	**21**	**21**	**21**	**21**	**21**	**21**	**21**	**21**	**21**	**21**	**21**
Debt service-to-revenue ratio											
Baseline	**31.9**	**10.9**	**17.7**	**11.6**	**12.5**	**13.5**	**13.8**	**12.7**	**11.5**	**10.3**	**19.0**
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2021-2041 1/	31.9	11.0	18.2	12.3	13.8	15.7	16.8	16.4	15.8	15.0	27.5
B. Bound Tests											
B1. Real GDP growth	31.9	11.8	20.8	13.7	14.7	16.0	16.3	15.0	13.6	12.1	22.4
B2. Primary balance	31.9	10.9	18.1	12.5	13.3	14.3	14.6	13.4	12.4	11.2	19.9
B3. Exports	31.9	11.0	18.3	13.1	13.8	14.8	15.0	13.8	13.2	12.9	21.4
B4. Other flows 2/	31.9	10.9	18.1	12.4	13.2	14.2	14.5	13.3	12.7	11.8	20.4
B6. One-time 30 percent nominal depreciation	31.9	13.8	22.4	14.1	15.2	16.6	17.1	15.7	14.2	11.8	23.0
B6. Combination of B1-B5	31.9	11.7	21.0	14.3	15.1	16.3	16.5	15.2	14.9	13.3	23.2
C. Tailored Tests											
C1. Combined contingent liabilities	31.9	10.9	18.1	12.1	12.9	14.0	14.3	13.2	12.0	10.6	19.3
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	31.9	10.9	17.8	12.7	14.7	15.4	14.4	15.4	10.6	10.0	18.8
Threshold	**23**	**23**	**23**	**23**	**23**	**23**	**23**	**23**	**23**	**23**	**23**

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.

2/ Includes official and private transfers and FDI.

Table 4. Rwanda: Sensitivity Analysis for Key Indicators of Public Debt, 2021–31

(In percent)

	Projections										
	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031
PV of Debt-to-GDP Ratio											
Baseline	**55.7**	**56.7**	**56.5**	**55.5**	**53.4**	**51.9**	**49.8**	**47.6**	**45.7**	**45.1**	**44.8**
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2021-2041 1/	56	56	56	56	56	55	54	53	53	52	53
B. Bound Tests											
B1. Real GDP growth	56	62	70	71	71	72	72	71	71	72	73
B2. Primary balance	56	59	61	60	58	56	54	51	49	48	48
B3. Exports	56	60	66	65	62	60	58	56	53	52	51
B4. Other flows 2/	56	60	62	61	59	57	55	52	50	49	48
B6. One-time 30 percent nominal depreciation	56	63	60	57	54	51	47	44	41	39	38
B6. Combination of B1-B5	56	57	60	59	57	56	54	53	51	50	50
C. Tailored Tests											
C1. Combined contingent liabilities	56	62	62	60	58	56	54	51	49	49	48
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	56	57	57	56	53	52	50	47	45	45	45
Public debt benchmark	**70**	**70**	**70**	**70**	**70**	**70**	**70**	**70**	**70**	**70**	**70**
PV of Debt-to-Revenue Ratio											
Baseline	**218.6**	**230.6**	**233.5**	**227.1**	**210.7**	**211.7**	**204.9**	**196.0**	**187.7**	**188.2**	**187.4**
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2021-2041 1/	219	227	230	227	215	221	219	216	212	216	217
B. Bound Tests											
B1. Real GDP growth	219	248	280	282	271	286	288	286	285	295	302
B2. Primary balance	219	238	254	246	228	228	221	211	202	202	200
B3. Exports	219	245	274	266	246	247	239	229	218	216	212
B4. Other flows 2/	219	244	258	250	232	233	225	216	206	205	202
B6. One-time 30 percent nominal depreciation	219	260	252	239	216	210	197	183	170	165	159
B6. Combination of B1-B5	219	231	245	241	225	228	223	216	208	210	210
C. Tailored Tests											
C1. Combined contingent liabilities	219	253	254	246	228	229	221	211	202	202	201
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	219	231	234	228	211	212	204	195	187	187	186
Debt Service-to-Revenue Ratio											
Baseline	**55.9**	**36.8**	**41.6**	**33.0**	**30.2**	**29.8**	**29.2**	**28.1**	**24.8**	**23.1**	**32.2**
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2021-2041 1/	56	37	43	35	32	32	32	31	28	26	37
B. Bound Tests											
B1. Real GDP growth	56	39	49	42	40	42	42	42	39	38	49
B2. Primary balance	56	37	44	37	33	33	31	30	26	25	34
B3. Exports	56	37	42	34	31	31	30	29	26	25	34
B4. Other flows 2/	56	37	42	34	31	30	30	29	26	24	33
B6. One-time 30 percent nominal depreciation	56	36	43	33	31	31	31	29	26	24	35
B6. Combination of B1-B5	56	36	44	36	33	33	33	32	28	26	36
C. Tailored Tests											
C1. Combined contingent liabilities	56	37	49	36	34	32	31	29	26	24	33
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	56	37	42	34	32	31	30	30	24	23	32

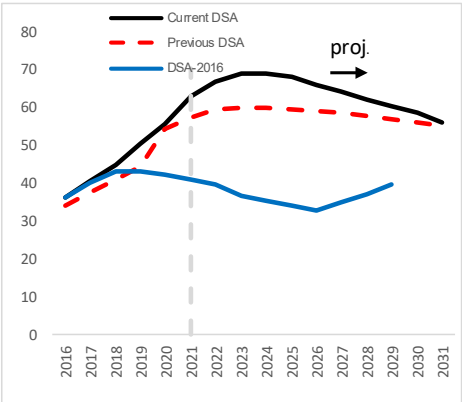
Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, GDP deflator and primary deficit in percent of GDP.

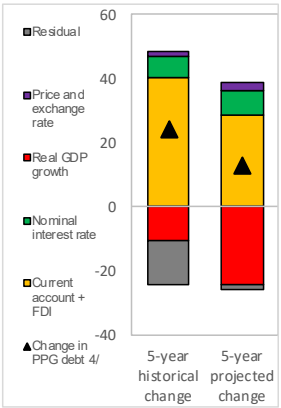
2/ Includes official and private transfers and FDI.

Figure 3. Rwanda: Drivers of Debt dynamics—Baseline Scenario[1/]

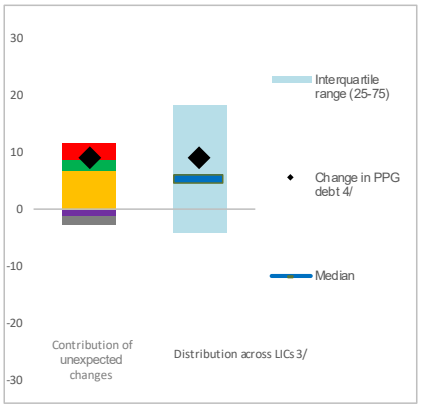
Gross Nominal PPG External Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)

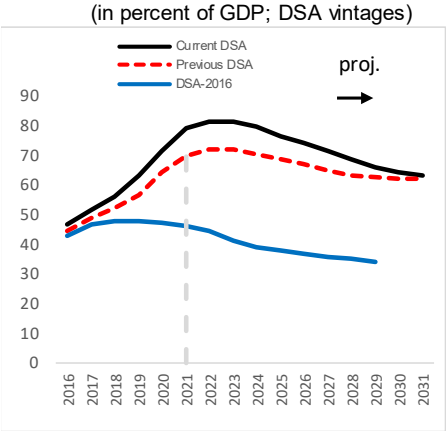


Unexpected Changes in Debt 2/
(past 5 years, percent of GDP)

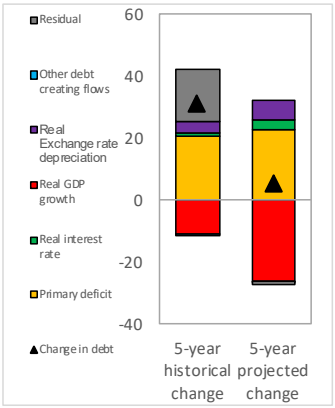


Public debt

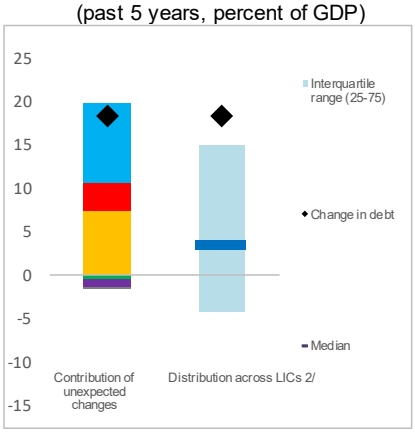
Gross Nominal Public Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)



Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)



1/ Compared to 2016 DSA and the previous full DSA in 2020.

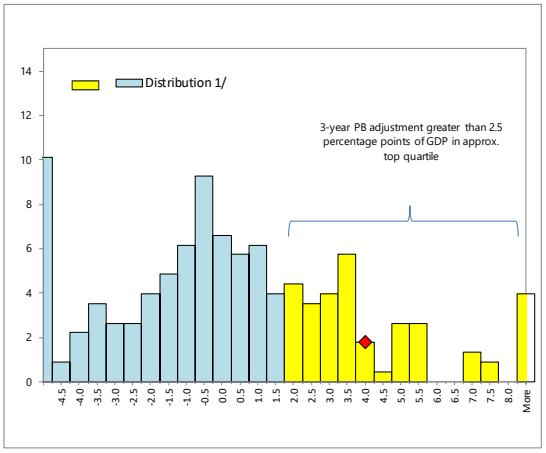
2/ Difference between anticipated and actual contributions on debt ratios.

3/ Distribution across LICs for which LIC DSAs were produced.

4/ Given the relatively low private external debt for average low-income countries, a ppt change in PPG external debt should be largely explained by the drivers of the external debt dynamics equation.

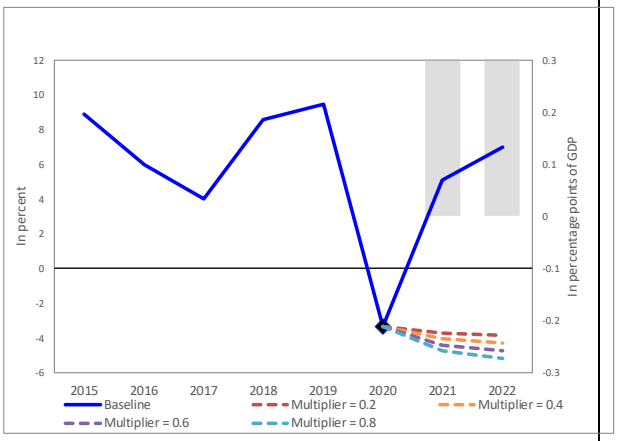
Figure 4. Rwanda: Realism Tools

3-Year Adjustment in Primary Balance (Percentage points of GDP)



1/ Data cover Fund-supported programs for LICs (excluding emergency financing) approved since 1990. The size of 3-year adjustment from program inception is found on the horizontal axis; the percent of sample is found on the vertical axis.

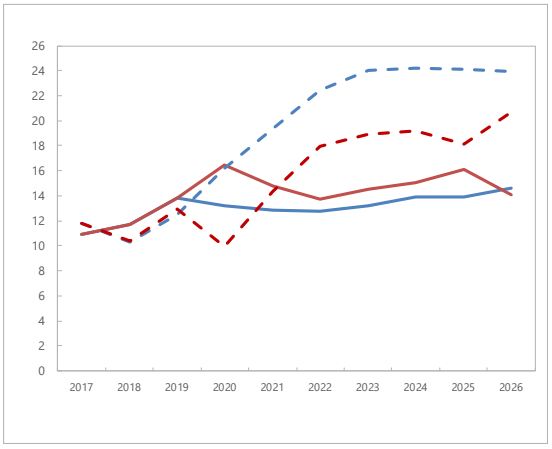
Fiscal Adjustment and Possible Growth Paths 1/



1/ Bars refer to annual projected fiscal adjustment (right-hand side scale) and lines show possible real GDP growth paths under different fiscal multipliers (left-hand side scale).

Public and Private Investment Rates 1/

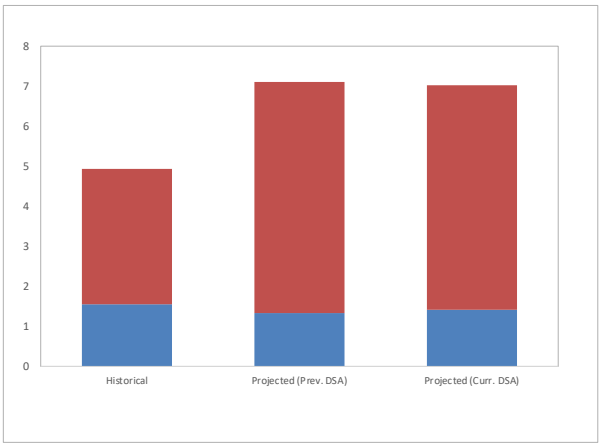
(% of GDP)



Gov. Invest. - Prev. DSA
Gov. Invest. - Current DSA
Priv. Invest. - Prev. DSA
Priv. Invest. - Current DSA

Contribution to Real GDP growth

(percent, 5-year average)



Contribution of other factors
Contribution of government capital

1/ Private investment includes the Bugesera airport project.

Figure 5. Rwanda: Qualification of the Moderate Category, 2021–31[1/]

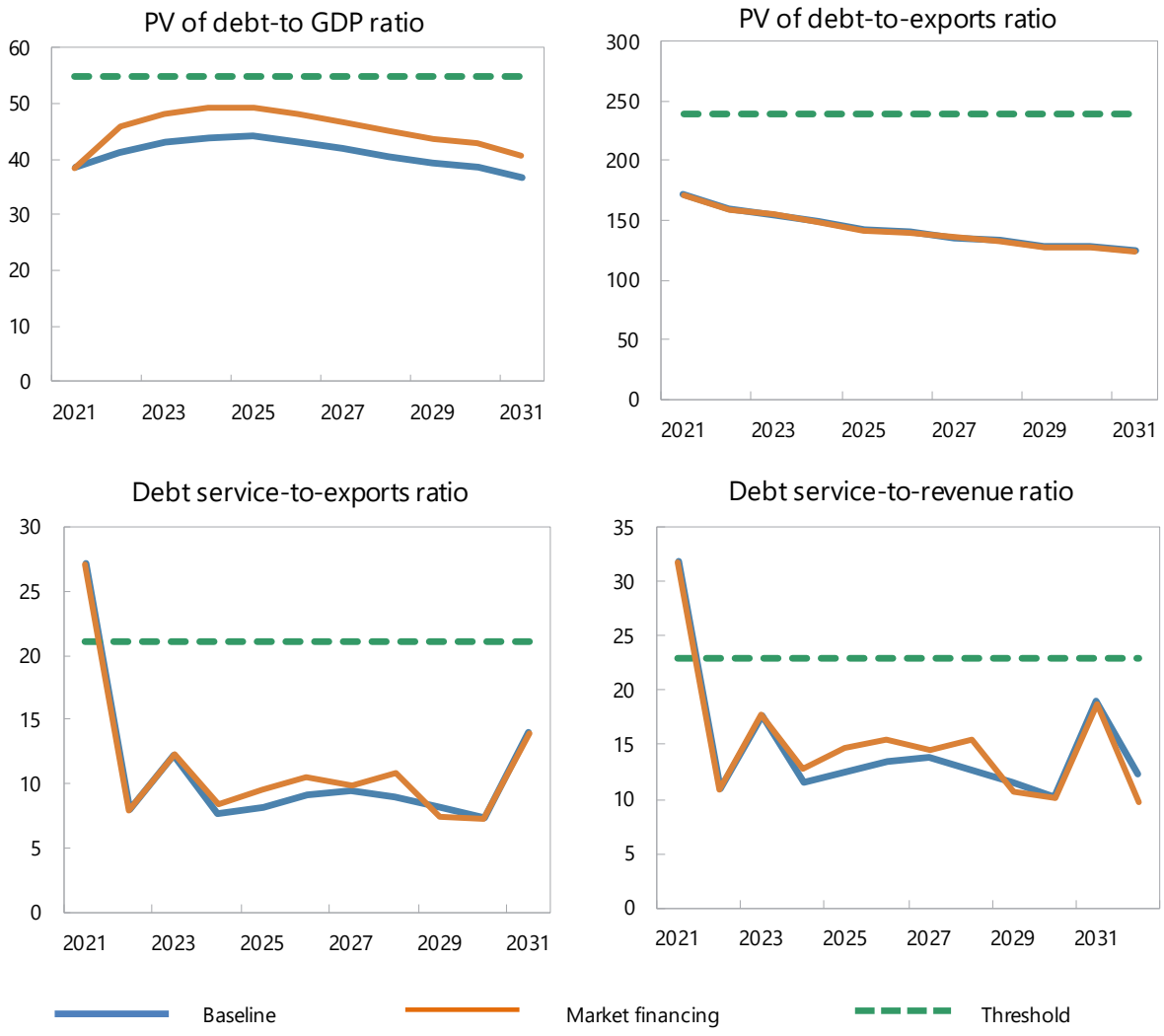
Sources: Country authorities; and staff estimates and projections.

1/ For the PV debt/GDP and PV debt/exports thresholds, x is 20 percent and y is 40 percent. For debt service/Exports and debt service/revenue thresholds, x is 12 percent and y is 35 percent.

Figure 6. Rwanda: Market-Financing Risk Indicators

	GFN 1/	EMBI 2/
Benchmarks	14	570
Values	21	n.a.
Breach of benchmark	Yes	n.a.
Potential heightened liquidity needs	High	

1/ Maximum gross financing needs (GFN) over 3-year baseline projection horizon.
2/ EMBI spreads correspond to the latest available data.



Sources: Country authorities; and staff estimates and projections.

Annex III. Financing Rwanda's Sustainable Development Goals

1. The pandemic has significantly raised Rwanda's spending needs to meet the Sustainable Development Goals (SDGs). Rwanda will need additional resources in the order of 21¼ percent of GDP per year to meet its SDGs by 2030.[1] This compares with a pre-pandemic SDG financing gap of 15.7 percent of GDP per year using the same model.[2] The additional needs could be 1–1½ percent of GDP larger if long-term scarring is considered. The widening of Rwanda's SDG financing gap post-pandemic can be mostly attributed to lower GDP (about 10 percent in real terms) and reduced fiscal space. The pandemic is expected to impose output losses to the tune of 10 percent in real terms, which translates into an additional SDG financing gap of 3.2 percent of GDP per year until 2030 (mostly reflecting higher infrastructure costs relative to the smaller size of the economy). Rwanda's commitment to safeguard debt sustainability entails a delay in fiscal adjustment that would require substantially reducing the country's overall fiscal deficit. The tighter stance reduces Rwanda's annual fiscal space to invest in SDGs, on average, by 2.4 percent of GDP over this decade.

Rwanda: Additional Grants Needed to Meet 2030 Target

	Percent of GDP		
	Pre-COVID	Post-COVID	Difference
Additional grants needed	15.7	21.3	5.6
of which: Health & Education	4.8	5.9	1.2
Infrastructure	11.0	15.4	4.4
Total change			**5.6**
Effect of lower nominal GDP			**3.2**
Effect of lower fiscal space			**2.4**
of which: change in revenues (-)			1.3
change in fiscal balance			2.2
change in interest expenses			0.5
change in non-SDG spending			-1.3
change in subsidies for private investment			0.0
change in identified grants (-)			-0.3

Source: IMF staff

2. Closing Rwanda's post-pandemic SDG financing gap will require ambitious fiscal measures to enhance domestic revenue mobilization and boost spending efficiency. Fiscal measures to enhance domestic revenue mobilization and boost spending efficiency have turned even more critical post-pandemic. By increasing government resources, they serve the twin purpose of closing SDG financing gaps and reducing public debt, consistent with Rwanda's fiscal consolidation objectives. Enacting a medium-term revenue strategy that gradually raises the total revenue-to-GDP ratio by about 7 percentage points above the baseline during 2023–29, reallocating one percentage point of GDP in public expenditures towards SDGs, and boosting spending efficiency to the level of peers could

Rwanda: Post-COVID Scenarios, 2020-2050

Scenario	SDGs met by	Additional grants per year [1]
A. Post-COVID baseline settings	>2050	0.0
B. Domestic revenue mobilization	2044	0.0
C. Spending reallocation to SDGs	2050	0.0
D. Combined fiscal measures	2041	0.0
E. Private sector participation	>2050	0.0
F. Active policies w/o addtl grants	2040	0.0
G. Baseline plus additional grants	2030	21.3
H. Fiscal measures plus grants	2030	15.6
I. Active policies plus grants	2030	13.7

Sources: IMF staff estimates and projections.
1 In percent of nominal GDP.

[1] Results are based on a dynamic financing framework developed in IMF Staff Discussion Note 2021/003 and applied for the case of Rwanda in Duarte Lledo and Perrelli (2021). Please refer to both papers for details.

[2] Estimates are based on the 3rd PCI review macroeconomic framework.

cover more than one quarter of the financing gap, helping the country to meet its SDGs around a decade later (by 2041).

3. Attracting private investment and additional resources to finance Rwanda's SDGs would also be critical. Increasing Rwanda's annual foreign direct investment to the top quartile of its peers' distribution could attract an extra 2¾ percent of GDP of private resources, closing up to one tenth of Rwanda's 2030 SDG financing gap. Polices to attract private investment should continue to (i) strengthen Rwanda's business climate, (ii) identify projects with an attractive risk-return profile through public-private partnerships, and (iii) grant government guarantees with due regard to fair and proportional distribution of risk and return among public and private participants.[3] Active policies that combine fiscal measures and higher private sector participation could fulfill more than one third of Rwanda's financing gap, enabling the country to meet its SDGs targets by 2040.

4. Financing Rwanda's SDGs cannot be achieved without additional support from development partners. Even in the scenario with active policies, the pledge to meet its SDGs by 2030 would still require about 13¾ percentage points of GDP in additional resources annually. Initiatives to leverage existing concessional resources to finance de-risking initiatives (blended finance), securing additional resources from established development partners, and unlocking untapped resources from institutional investors (e.g. pension and mutual funds, insurance companies) and private philanthropies must be vigorously pursued.[4]

[3] IMF Department Paper 21/11.

[4] Ibid.